Days 1–5: Build and Configure a Small Office Network

Network Setup and Configuration Report

1. Introduction and Objectives

This report describes a small office LAN design using static IP addressing, DHCP, DNS, VLAN segmentation, a router, and NAT. The network can be built in Cisco Packet Tracer or another approved lab environment. The purpose is to practise configuration and document the tests used to verify the network.

Objectives:

- Assign IP addresses and confirm connectivity between devices in the same subnet.
- Provide automatic IP configuration using DHCP.
- Resolve a custom hostname using a local DNS service or a clearly documented hosts-file test.
- Separate Workstations and Smart Devices into two VLANs.
- Configure router NAT and test access to an external or simulated server.
- Record troubleshooting steps and actual test results.

2. Proposed Topology

Suggested lab topology: five PCs/laptops and two smart-device test clients connect to a managed switch. The switch connects to a router. A server may provide DHCP, DNS, and a local web page. The router connects to a simulated ISP/external server or an approved lab internet connection.

Topology sketch (logical):

PC1, PC2, PC3, PC4, PC5, Smart1, Smart2 → Managed Switch → Router → ISP / External Server
Server (DHCP/DNS/Web, as supported by the lab)

Screenshot placeholder: Insert your Packet Tracer/GNS3 topology screenshot here.

3. IP Addressing and Subnet Plan

Example lab plan: use private IPv4 addresses. VLAN 10 and VLAN 20 use separate /24 networks, so devices in different VLANs are in different subnets. The IPs below are a proposed plan, not proof that the lab has been configured.

Device / role	VLAN	Example IP / range	Subnet mask	Default gateway
Router VLAN 10 gateway	10	192.168.10.1	255.255.255.0	—
PC1 (static)	10	192.168.10.11	255.255.255.0	192.168.10.1
PC2 (static)	10	192.168.10.12	255.255.255.0	192.168.10.1
PC3 (static)	10	192.168.10.13	255.255.255.0	192.168.10.1
PC4 (static)	10	192.168.10.14	255.255.255.0	192.168.10.1
PC5 (static)	10	192.168.10.15	255.255.255.0	192.168.10.1
DHCP client pool (example)	10	192.168.10.100–192.168.10.149	255.255.255.0	192.168.10.1
Router VLAN 20 gateway	20	192.168.20.1	255.255.255.0	—
Smart1	20	192.168.20.11 or DHCP	255.255.255.0	192.168.20.1
Smart2	20	192.168.20.12 or DHCP	255.255.255.0	192.168.20.1
Local DNS/web server (if used)	10	192.168.10.50	255.255.255.0	192.168.10.1

Important: Exclude static addresses from the DHCP pool to prevent duplicate IP addresses. A /24 network has 254 normally usable IPv4 host addresses. If the router or server uses different addresses, update the plan consistently.

4. Task 1 — Static IPs and Ping Verification

1. Choose five devices in the lab and assign the static IP addresses from the table. Enter the subnet mask and default gateway as well.
2. Ensure the five devices are connected to the same VLAN/subnet for the same-subnet ping test.
3. On Windows, open Command Prompt and run ipconfig to verify the configuration.

4. From PC1, run ping 192.168.10.12, ping 192.168.10.13, ping 192.168.10.14, and ping 192.168.10.15. Use the actual addresses configured in your lab.
5. Record replies or errors. If ping fails, check IP address, subnet mask, cable/link, VLAN assignment, and host firewall.

5. Task 2 — DHCP Server and Lease Verification

6. Enable DHCP on the router or lab server, depending on the simulator features.
7. Create a pool for the intended subnet. Set the network/subnet mask, default gateway, DNS server, and a range of at least three addresses.
8. Keep static addresses and server/gateway addresses outside the dynamic pool or configure exclusions.
9. Set at least three client devices to obtain their IP configuration automatically. In Packet Tracer PC, use Desktop > IP Configuration > DHCP.
10. Check each client with ipconfig /all (Windows) or the simulator's IP configuration window. Verify that addresses are unique and belong to the correct pool.

Troubleshooting: If a client has a 169.254.x.x address or no lease, check link/VLAN, DHCP service, pool availability, scope settings, and DHCP relay for another subnet.

Screenshot placeholder: DHCP pool configuration and three clients' assigned addresses.

6. Task 3 — Local DNS and Custom Hostname

Example hostname: musicplayer.local. Configure the lab DNS service with an A record that maps this name to the local web server IP, for example 192.168.10.50. The server must actually host a web page, and clients must use the DNS server that contains the record. The .local suffix can conflict with multicast DNS on some systems; for a controlled lab, use the simulator's supported name-resolution feature or a lab-only domain such as musicplayer.test.

11. Set the local web server IP to the planned static address and enable its HTTP service if supported.
12. In the DNS service, add an A record: musicplayer.test → 192.168.10.50 (replace with your actual server IP).
13. Set the DHCP DNS option or client DNS setting to the address of the DNS server.
14. From a client, run nslookup musicplayer.test. Confirm that it returns the intended server IP.
15. Open http://musicplayer.test in the browser if the simulator supports DNS-based browser access. If using a hosts-file test instead, document that it is a local workaround and not central DNS.

7. Task 4 — VLAN 10 and VLAN 20 Segmentation

Create VLAN 10 named Workstations and VLAN 20 named SmartDevices on a VLAN-capable switch. Put at least two end devices in each VLAN. A VLAN separates Layer 2 broadcast domains. Devices in different VLANs normally cannot communicate directly unless inter-VLAN routing is configured and permitted by access rules.

Example Cisco IOS commands (adjust interface numbers to match your topology):

```
enable
configure terminal
vlan 10
 name Workstations
vlan 20
 name SmartDevices
!
interface range fastEthernet 0/1-2
 switchport mode access
 switchport access vlan 10
!
interface range fastEthernet 0/3-4
 switchport mode access
 switchport access vlan 20
!
```

```
end
show vlan brief
```

For the test, assign two clients to each VLAN and use IP addresses from the matching subnet. Ping between devices in the same VLAN; then ping across VLANs. If there is no inter-VLAN routing, cross-VLAN pings should fail. If routing is configured, use ACLs or the lab's security policy to block the traffic if the assignment requires isolation.

Connectivity test	Expected result in isolated VLAN design	Actual result
VLAN 10 device A → VLAN 10 device B	Success	_____
VLAN 20 device A → VLAN 20 device B	Success	_____
VLAN 10 device → VLAN 20 device	Fail if inter-VLAN routing is not enabled/allowed	_____

Verification command: show vlan brief. Screenshot placeholder: VLAN membership and ping tests.

8. Task 5 — Router NAT and Simulated Internet Access

NAT/PAT translates private LAN source addresses to an address on the router's outside interface. A real internet test requires a working upstream connection; in a simulator, use an ISP router and an external server to represent the Internet.

Example Cisco IOS NAT/PAT template (interface names and addresses must match the actual lab):

```
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
!
interface <LAN-interface-or-subinterfaces>
 ip nat inside
!
interface <WAN-interface>
 ip nat outside
!
ip nat inside source list 1 interface <WAN-interface> overload
```

Replace placeholders with the correct interface names. Configure the WAN address, upstream/default route, and return path as required by the simulator. For router-on-a-stick, mark each relevant LAN subinterface as ip nat inside. Do not paste the template without adapting it to the topology.

16. From a LAN client, first ping its default gateway.
17. Ping the simulated external server IP address. If this fails, check routing, WAN link, ACLs, and NAT configuration.
18. If external IP connectivity works but a domain name does not, check DNS separately.
19. On a Cisco router, use show ip nat translations and show ip nat statistics to inspect NAT activity.
20. Record the actual result. A Packet Tracer simulation does not automatically provide unrestricted public Internet access.

9. Troubleshooting Summary

Problem	Possible cause	Checks	Solution
Same-subnet ping fails	Wrong IP/mask, link down, firewall, wrong VLAN	ipconfig, cable/link, show vlan brief	Correct IP/mask, restore link, correct VLAN or firewall rule
Client has no DHCP lease	DHCP service/pool issue, VLAN/relay issue	ipconfig /all, DHCP pool and service status	Fix scope, service, VLAN or relay; renew lease
Hostname does not resolve	Wrong DNS server or missing record	nslookup, client DNS settings, DNS records	Correct DNS option/record and clear cache
Cross-VLAN traffic unexpectedly works	Inter-VLAN routing allows it	Check router ACLs and VLAN interfaces	Apply intended ACL/security policy
No external access	Missing default route, NAT, WAN link or DNS	Ping gateway, external IP, then hostname; show ip nat statistics	Correct route/NAT/WAN/DNS as indicated

10. Conclusion

A small office network needs a consistent IP plan, working DHCP and DNS, correct VLAN membership, and a router with suitable routing and NAT settings. Each configuration should be verified with commands or simulator tests, and actual evidence should be included in the final report.

Final Submission Checklist

☐ Five static IP devices configured and ping results recorded

☐ Three DHCP clients verified with unique leases

☐ Custom hostname resolves and local page test documented

☐ VLAN 10 and VLAN 20 created with at least two devices each

☐ Same-VLAN and cross-VLAN tests recorded

☐ NAT and simulated external access test documented

☐ Actual screenshots inserted and blank result fields completed

Academic honesty note: This document is a report template with an example IP plan and command templates. Fill in actual device names, addresses, outputs, screenshots, and test results after completing the lab. Do not state that a configuration succeeded unless you verified it.